Task-04: Customer Churn Prediction Using Classification Models

In this task, I was asked to implement and evaluate the performances of **various classification models** on the **Telco Customer Churn Dataset.**The main goal was to predict customer churn based on the other features. Before implementing the model, various other steps were involved. These steps include:

- ☐ Importing the standard libraries and modules
- ☐ Getting the dataset
- ☐ Exploratory data analysis
- ☐ Splitting the data into test and train sets
- ☐ Transformation of categorical columns into numerical columns

For this study, two classification models were used: logistic regression and decision tree classifier. After performing all the data preprocessing steps, these models were trained on the transformed data and their predictions were recorded. Their performances were measured using the standard metrics such as Accuracy, Precision, Recall, F1 Score, and AUC score. Confusion matrix, ROC plot, and classification reports were also created.

METHODOLOGY

1. Exploratory Data Analysis:

Exploratory data analysis is the process of analysing, summarizing, and visualizing the dataset to understand the structure and spot anomalies before feeding the data to the machine learning models. In this task, I performed EDA in multiple steps:

1. Inspecting the dataset by checking the columns to know more about the dataset's features and its types and the range of values present.

2. Checking for missing values to either impute or drop such values

3. Calculating summarised statistics of the features to get a summary of the features

4. Visualising the distribution of features using histograms.

2. Train-test split and Column Transformation:

After performing EDA, the data was split into training and test sets using traintestsplit. It was split into an 80:20 ratio. The dataset contained both numerical and categorical features(both in numeric and non-numeric form). A pipeline was created through which the numeric columns were imputed and standardized, while the categorical columns were encoded into

binary numerics using OneHotEncoder. This pipeline was applied to the training and test sets to obtain transformed sets, ready to be used for training models.

3. Model Training and Evaluation

The two models (logistic regression, decision tree) were trained on the training features and labels, and predictions were obtained on the basis of test features. Logistic regression was tuned again to handle imbalance classes. All three models were trained and evaluated on various metrics.

The following evaluation metrics were used, along with the reason why they were selected:

- **Accuracy**: Measures the overall proportion of correctly classified customers.
- Precision: Measures how many customers predicted to churn actually churned, helping assess false positive predictions.
- **Recall**: Measures how many actual churn cases were correctly identified.
- **F1-Score**: Provides a balance between Precision and Recall, making it useful for evaluating performance on an imbalanced dataset.
- **ROC-AUC Score**: Measures the model's ability to distinguish between churn and non-churn customers across different classification thresholds.
- **Confusion Matrix** : Provides a detailed breakdown of true positives, true negatives, false positives, and false negatives, helping identify the types of classification errors made by each model.
- **ROC Plot**: Visually compares the True Positive Rate against the False Positive Rate at different classification thresholds, helping assess the models' ability to distinguish between the two classes.
- **Classification Report**: Summarizes Precision, Recall, F1-score, and support for each class, providing a detailed view of model performance for both churn and non-churn customers.

4. Class Imbalance Analysis

Class imbalance means that one class is present in a much larger number in comparison to the other classes. In the selected dataset, almost 73% of the datapoints in the Churn column contained “No”, which is about three times the quantity of “Yes” points. This makes this dataset imbalance. To handle imbalancing, logistic regression was fine-tuned to perform class weighting.

RESULTS

The following results were obtained after performance evaluation: Model Performance Comparison Table

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.821150	0.686154	0.597855	0.638968	0.862211
1	Decision Tree	0.799858	0.618799	0.635389	0.626984	0.840000
2	Tuned Logistic Regression	0.749468	0.516779	0.825737	0.635707	0.862040

Classification Report of Logistic Regression:

	precision	recall	f1-score	support
Churn	0.86	0.90	0.88	1036
Yes	0.69	0.60	0.64	373
accuracy			0.82	1409
macro avg	0.77	0.75	0.76	1409
weighted avg	0.82	0.82	0.82	1409

Classification Report of Decision Tree:

	precision	recall	f1-score	support
0	0.87	0.86	0.86	1036
1	0.62	0.64	0.63	373
accuracy			0.80	1409
macro avg	0.74	0.75	0.75	1409
weighted avg	0.80	0.80	0.80	1409

Classification Report of Tuned Logistic Regression:

	precision	recall	f1-score	support
0	0.92	0.72	0.81	1036
1	0.52	0.83	0.64	373
accuracy			0.75	1409
macro avg	0.72	0.77	0.72	1409
weighted avg	0.81	0.75	0.76	1409

DISCUSSION

Final Model Selection

The tuned logistic regression model would be our selected model for this dataset. The reasons are as as follows:

- ➢ Highest accuracy: 82.11%, meaning it correctly classifies the largest proportion of customers.
- ➢ Highest precision: 68.62%, so its churn predictions are more reliable than those of the other two models.
- ➢ Highest F1-score: 63.90%, although only marginally higher than the balanced model's 63.57%.
- ➢ Highest ROC-AUC: 86.22%, indicating strong ability to distinguish between churn and non-churn customers across classification thresholds.

CONCLUSION

In this project, a standard machine learning pipeline was followed. This includes importing the dataset, exploratory data analysis, feature scaling, train-test split, column transformation, model training, obtaining predictions, and evaluation of models on the basis of standard classification metrics.

After performing each test carefully, it is concluded that the fine-tuned logistic regression was the best performer amongst the models, as it had the highest score on every evaluation metrics used.